

# Simon Hathout Willard

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Third year Engineering Physics student at UBC with skills in CAD, software development, and engineering design. Strong computer skills and able to pick up new software and tools quickly.

## TECHNICAL EXPERIENCE

### **Automation Engineering Intern at Dometic Vancouver | January – April 2025**

- Designed and implemented SCADA software using the Ignition suite, reducing scrap product and improving machine scheduling, leading to savings of \$15000 in 4 months.
- Created and lead trainings on SCADA software operation, ensuring buy-in from engineering and operations teams.
- Wrote Python code to process data and create interactive Human Machine Interfaces through Ignition's Vision and Perspective modules.
- Used SQL (SQLserver) to store machine data effectively, preserve ACID principles, and enforce relevant constraints on data.

### **Automation Subteam Member with Agroponics Design Team | 2023 - 2025**

- Used IOT sensor systems with Arduino and ESP32 to measure PH and temperature in a nutrient solution, improving plant growth and productivity.
- Communicated with other subteams to determine required data and assisted in selection and procurement of appropriate sensors.

### **NorthForge Design Space Member | July-August 2024**

- Received formal training with woodshop and laser cutter tools, ensuring safe use.
- Worked part time with Maker Mind Toys, making wooden toys for children.

## SKILLS

**Programming** - Java, Python, SQL, C++, Matlab, Arduino

**Design Tools** - Fusion 360, Onshape, Machine Shop, 3D Printing, Laser Cutting, Woodshop

**Software** - Git, GitHub, Linux OS. Microsoft and Google suites

## EDUCATION

### **3<sup>rd</sup> Year Engineering Physics | 2023 – Present | UBC Applied Sciences**

- Courses include software development, mechanical and electrical design, thermodynamics, and machine learning. A full list can be found at <https://hathoutwillard.github.io/courses/>
- Worked in a team of 4 to design and fabricate a robot with a three-axis arm, to autonomously navigate a course and retrieve objectives.
- Teaching assistant for CPEN 221 (Software Development). Led a lab section, answered discussion board questions, and assisted with grading.
- Cumulative average: 93% (A+)